

Problem 14 - Rate of Change of $\vec{H}_{/c}$

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(Pg 1)

>> PROBLEM STATEMENT:

Candidate definitions for $\vec{H}_{/c}$ - for which is $\vec{M}_{/c} = \frac{d}{dt} \vec{H}_{/c}$ true?

$$(a) \vec{H}_{/c}^A = \sum \vec{r}_{i/c'} \times \vec{v}_{i/c'} m_i$$

Where c' is a point fixed in $\mathcal{F}$ that instantaneously coincides with c .

$$(b) \vec{H}_{/c}^B = \sum \vec{r}_{i/c} \times \vec{v}_{i/o} m_i$$

$$(c) \vec{H}_{/c}^C = \sum \vec{r}_{i/c} \times \vec{v}_{i/c} m_i$$

How do we figure this out?

BASIC FACT:

(Core of Problem, take as given)

$$\sum \vec{M}_c^{\text{ext}} = \sum_{\text{whole system}} \vec{r}_{i/c} \times m_i \vec{a}_i$$

why? Various Explanations

- (a) internal moments cancel (assume)
- (b) pairwise equal & opposite forces
- (c) internal forces do no work in rigid rotations

Basic Test:

$$\text{Does } \frac{d}{dt} \vec{H}_{/c} = \sum \vec{r}_{i/c} \times m_i \vec{a}_i \quad ??$$

if so:
 $\vec{H}_{/c}$ is good
because
 $\frac{d}{dt} \vec{H}_{/c} = \vec{M}_c$

if not:
 $\vec{H}_{/c}$ is NOT good
(surprise surprise)

OK, Now Let's try it out on each definition.

→ TEST (a):

$$\vec{H}_{/c}^A = \sum \vec{r}_{i/c'} \times \vec{v}_{i/c'} m_i \quad (\text{Here's our CANDIDATE ... will it pass the test?})$$

will $\frac{d}{dt} \vec{H}_{/c} = \vec{M}_{/c}$?

$$\Downarrow \frac{d}{dt} \vec{H}_{/c}^A = \sum (\underbrace{\vec{r}_{i/c'}}_{=\vec{r}_{i/c}} \times \underbrace{\vec{a}_{i/c'}}_{=\vec{a}_{i/o}} + \underbrace{\vec{v}_{i/c'} \times \vec{v}_{i/c'}}_{=0}) m_i$$

(since c' is
coincident
with c)

(since c'
is fixed
in $\mathcal{F}$)

(since $\vec{v}_{i/c'}$ is
necessarily parallel
to itself)

YES!

$$\therefore \frac{d}{dt} \vec{H}_{/c}^A = \frac{d}{dt} \left(\sum \vec{r}_{i/c'} \times \vec{v}_{i/c'} m_i \right) = \sum \vec{r}_{i/c} \times \vec{a}_{i/o} m_i = \vec{M}_{/c}$$

THE DEFINITION WORKS IN GENERAL

→ TEST (b):

$$\vec{H}_{/c}^B = \sum \vec{r}_{i/c} \times \vec{v}_{i/o} m_i \quad (\text{Our new CANDIDATE ... will it pass the test?})$$

will $\frac{d}{dt} \vec{H}_{/c} = \vec{M}_{/c}$?

$$\Downarrow \frac{d}{dt} \vec{H}_{/c}^B = \sum (\vec{r}_{i/c} \times \vec{a}_{i/o} + \vec{v}_{i/c} \times \vec{v}_{i/o}) m_i$$

... it depends. In
certain conditions it
works.

If $\sum (\vec{v}_{i/c} \times \vec{v}_{i/o}) m_i = 0$ THEN
the definition will work.

This happens if:

c is any point whose
velocity is parallel to
the velocity of the COM.
(including the case where
 c is the COM)

Proof:

$$\sum (\vec{v}_{i/c} \times \vec{v}_{i/o}) m_i \Rightarrow \sum (\vec{v}_{i/o} - \vec{v}_{c/o}) \times \vec{v}_{i/o} m_i \Rightarrow \sum \vec{v}_{i/o} \times \vec{v}_{i/o} - \vec{v}_{c/o} \times \sum \vec{v}_{i/o} m_i$$

$$\sum (\vec{v}_{i/o} \times \vec{v}_{i/o} + \vec{v}_{c/o} \times \vec{v}_{i/o}) m_i \leftarrow \sum \vec{v}_{c/o} \times (\vec{v}_{i/o} + \vec{v}_{c/o}) m_i$$

$$\vec{v}_{c/o} \times \sum \vec{v}_{i/o} m_i = 0 \Rightarrow \vec{v}_{c/o} \times \vec{v}_{G/o} m_{TOT} = 0$$

Not dependent on i ,
OK to take
outside of sum.

if $\vec{v}_{c/o} \parallel \vec{v}_{G/o}$

$$\text{So } \frac{d}{dt} \vec{H}_{/c}^B = \sum (\vec{r}_{i/c} \times \vec{a}_{i/o} + \vec{v}_{c/o} \times \vec{v}_{G/o} m_{TOT}) m_i \Rightarrow \text{So } \frac{d}{dt} \vec{H}_{/c}^B = \vec{M}_{/c} \quad \text{YAY!}$$

c is any fixed point.

Proof:

$$\vec{v}_{i/c} = \vec{v}_{i/o} \quad \text{for fixed } c$$

$$\frac{d}{dt} \vec{H}_{/c}^B = \sum (\vec{r}_{i/c} \times \vec{a}_{i/o} + \vec{v}_{i/c} \times \vec{v}_{i/o}) m_i$$

$$\text{So } \frac{d}{dt} \vec{H}_{/c}^B = \vec{M}_{/c} \quad 0$$

YAY!

→ TEST (c):

$$\vec{H}_{/c}^c = \sum (\vec{r}_{i/c} \times \vec{v}_{i/c}) m_i$$

(Here's our CANDIDATE ... will it pass the test?)

will $\frac{d}{dt} \vec{H}_{/c}^c = \vec{M}_{/c}$?

$$\frac{d}{dt} \vec{H}_{/c}^c = \sum (\vec{r}_{i/c} \times \vec{a}_{i/c} + \vec{v}_{i/c} \times \vec{v}_{i/c}) m_i$$

$$\frac{d}{dt} \vec{H}_{/c}^c = \sum \vec{r}_{i/c} \times \vec{a}_{i/c} m_i$$

... it depends. In certain conditions it works.

If $\sum (\vec{r}_{i/c} \times \vec{a}_{i/c}) m_i = \sum (\vec{r}_{i/c} \times \vec{a}_{i/o}) m_i$ THEN the definition works.

Let's manipulate the equation a bit more so we can easily see when the above is true.

What we want to do is separate out the part we are interested in: $\sum (\vec{r}_{i/c} \times \vec{a}_{i/o}) m_i$ from what we found: $\sum (\vec{r}_{i/c} \times \vec{a}_{i/c}) m_i$.

because that's $\vec{M}_{/c}$ because that's $\frac{d}{dt} \vec{H}_{/c}^c$

THEN we can see which conditions the definition works in: any for which the leftover part = 0

So... when is this zero?

$$\vec{r}_{G/c} \parallel \vec{a}_{c/o}$$

Proof:

remember I want to prove $\sum \vec{r}_{i/c} \times \vec{a}_{c/o} m_i = 0$ if $\vec{r}_{G/c} \parallel \vec{a}_{c/o}$.

$$\sum m_i (\vec{r}_{i/o} - \vec{r}_{c/o}) \times \vec{a}_{c/o}$$

$$(\sum m_i \vec{r}_{i/o} - \sum m_i \vec{r}_{c/o}) \times \vec{a}_{c/o}$$

$$(m_{TOT} \vec{r}_G - m_{TOT} \vec{r}_{c/o}) \times \vec{a}_{c/o}$$

$$m_{TOT} \vec{r}_{G/c} \times \vec{a}_{c/o} \rightarrow \text{which is 0 if } \vec{r}_{G/c} \parallel \vec{a}_{c/o}$$

$$\rightarrow \text{So: } \frac{d}{dt} \vec{H}_{/c}^c = \vec{M}_{/c} \text{ YAY!}$$

C is COM

Proof:

(c = G so I can switch subscripts as I please)

Remember I want to prove $\sum (\vec{r}_{i/c} \times \vec{a}_{c/o}) m_i = 0$ if c = G

$$\sum \vec{r}_{i/G} \times \vec{a}_{G/o} m_i$$

switch c w/ G

$$(\sum \vec{r}_{i/G} m_i) \times \vec{a}_{G/o}$$

take out of $\sum$ bcs not dependent

$$\sum \vec{r}_i m_i - \sum \vec{r}_G m_i = m_{TOT} \vec{r}_G - m_{TOT} \vec{r}_G = 0$$

thus for c = G $\frac{d}{dt} \vec{H}_{/c}^c = \sum (\vec{r}_{i/c} \times \vec{a}_{i/o}) m_i$

$$\rightarrow \text{So: } \frac{d}{dt} \vec{H}_{/c}^c = \vec{M}_{/c} \text{ YAY!}$$

$$\vec{a}_{c/o} = 0$$

(i.e. C is a fixed point OR C moves at const. vel.)

Proof:

$$\vec{a}_{c/o} = 0$$

$$\frac{d}{dt} \vec{H}_{/c}^c = \sum (\vec{r}_{i/c} \times \vec{a}_{i/o}) m_i - \sum \vec{r}_{i/c} \times \vec{a}_{c/o} m_i$$

→ So:

$$\frac{d}{dt} \vec{H}_{/c}^c = \vec{M}_{/c}$$

YAY!